

TravelLink Backend — Progress Notes

Project Location

F:\Projects\TravelLink\Server\Server\

Tech Stack

- .NET 8, ASP.NET Core Web API
 - Entity Framework Core + SQL Server (TravelLinkDb)
 - JWT Authentication (Microsoft.AspNetCore.Authentication.JwtBearer v8.0.0)
 - BCrypt.Net-Next (password hashing)
 - CloudinaryDotNet (image storage)
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☐ Completed Modules

Entities (Models/Entities/)

- User.cs — includes ImageUrl, CreatedGroups nav property
- FriendRequest.cs
- Group.cs — includes CoverImageUrl
- GroupMember.cs
- Expense.cs
- ExpenseSplit.cs

AppDbContext (Data/AppDbContext.cs)

- All 6 DbSet's registered
- Full Fluent API configured (dual FK Restrict on FriendRequest, Cascade on Group/Expense/ExpenseSplit)

Database

- appsettings.json — connection string + Jwt section + Cloudinary section all correct
- InitialCreate migration applied — TravelLinkDb live with all 6 tables

Program.cs

- DbContext, JWT, CORS (app.UseCors("auth") applied), Cloudinary Singleton
- All services registered: IAuthService, IFriendService, IGroupService,

IImageService, IExpenseService

- UseAuthentication() + UseAuthorization() in correct order

Auth Module ☐

- Register / Login endpoints tested & working

Friend System Module ☐

Endpoint	Route
Send request	POST /api/friend/send
Accept/Reject	POST /api/friend/respond/{id}? action=accept\ reject
Pending requests	GET /api/friend/pending
Friends list	GET /api/friend/list

Group Module ☐

- Auto-deletes group when last member leaves
- Admin transfer when admin leaves and no other admin exists | Endpoint | Route | |---|---| |
Create group | POST /api/group/create | | My groups | GET /api/group/my | |
Get group by ID | GET /api/group/{id} | | Add member (Admin) | POST
/api/group/{id}/add-member | | Leave group | DELETE
/api/group/{id}/leave | | Remove member (Admin) | DELETE
/api/group/{id}/remove-member | | Upload group cover (Admin) | POST
/api/group/{id:guid}/cover |

Image Module ☐ (Cloudinary)

- IImageService / ImageService (UploadImageAsync, DeleteImageAsync)
- Bugfix: strips version segment from URL before extracting public ID for deletion

User Module ☐

Endpoint	Route
Get profile	GET /api/user/profile
Upload/Update avatar	POST /api/user/avatar

☐ Expense Module — COMPLETE

Design Decisions Made

- `SplitType`: "Equal" or "Custom" — handled in service layer, no DB column needed
- `ParticipantIds` field added to `CreateExpenseDto` for 1-on-1 equal splits (no `GroupId`)
- Payer NOT force-added to splits — supports "I paid for others, not myself" scenario
- `Warning` field added to `ExpenseDto` to notify payer when they're not in their own split
- Rounding remainder assigned to last participant (e.g. ₹100 ÷ 3 = 33.33 + 33.33 + 33.34)
- Payer's own split auto-marked `IsPaid = true` at creation
- Delete blocked if any other participant has already settled (protects financial records)
- Balance netting algorithm: eliminates mutual debts (A → B ₹150, B → A ₹100 → A → B ₹50)

DTOs (DT0s/Expenses/) ☐

- `CreateExpenseDto.cs` — includes `ParticipantIds`, `SplitType`, `Splits`
- `ExpenseSplitInputDto.cs`
- `ExpenseSplitDto.cs`
- `ExpenseDto.cs` — includes `Warning?` field
- `BalancedDto.cs`
- `GroupAnalyticsDto.cs` — nested `CategoryBreakdownDto`, `MemberContributionDto` (lists initialized)
- `UserToUserDto.cs` — nested `UserTransactionDto` (typo `TotalYouOuw` → `TotalYouOwe` fixed)

Services ☐

- `IExpenseService.cs`
- `ExpenseService.cs` — all 8 methods implemented

Controller ☐

Endpoint	Route
Add expense	POST /api/expense
Get expense by ID	GET /api/expense/{id}
Delete expense	DELETE /api/expense/{id}
Get group expenses	GET /api/expense/group/{groupId}
Get group balances	GET

Endpoint	Route
	/api/expense/group/{groupId}/balances
Get group analytics	GET /api/expense/group/{groupId}/analytics
Settle a split	PUT /api/expense/split/{splitId}/pay
User-to-user dashboard	GET /api/expense/user/{userId}/summary

Bugs Fixed During This Session

#	File	Bug
1	ExpenseService.cs	Parameter order mismatch in GetUserToUserSummaryAsync
2	ExpenseService.cs	NullReferenceException in MapToExpenseDto (PaidBy?.)
3	ExpenseService.cs	return null → Task.FromResult<>(null) in non-async method
4	ExpenseService.cs	Missing semicolon on Warning assignment
5	ExpenseService.cs	Force-added payer to equal splits (removed)
6	ExpenseService.cs	Non-nullable return types returning null
7	Program.cs	app.UseCors("auth") was registered but never applied
8	Program.cs	Duplicate AddSwaggerGen() call removed
9	GroupController.cs	Missing :guid constraint on cover route
10	GroupController.cs	Used raw Guid.Parse instead of GetUserId() helper
11	GroupAnalyticsDto.cs	Lists not initialized; strings not initialized

#	File	Bug
12	FriendController.cs	Typo requests
13	User.cs	Email not initialized

➔ Remaining Modules

1. Real-Time Location Tracking (SignalR)

- SignalR hub setup
- LocationHub.cs — broadcast user coordinates to group members
- Endpoints: join/leave location session, broadcast position

2. AI Itinerary Builder (Gemini API)

- Gemini API integration
- Prompt engineering for trip itinerary generation
- Endpoint: POST /api/itinerary/generate